

Air-to-Air Refuelling Academy — Boom and Receptacle (USAF)

F-16C-50 · Viper · KC-135 Stratotanker (boom) · Modern (2000s+)

Eleven short rides that teach flying-boom refuelling the way it is actually taught — formation first, plug last. You start in position every time; the commute is deleted until ride 7, where it comes back on purpose. Ten of the eleven are graded in-mission — every ride except the operational capstone.

The receiver holds still and a boom operator flies the boom into the receptacle. Your entire job is to stop moving. That is a formation problem wearing a refuelling costume, and this track trains it in that order: hold a position, control closure, then plug — not the reverse.

How to use this

Fly the rides **in order**. Each one adds exactly one new demand, and the order is the point — the commonest way people fail to learn air refuelling is starting at the plug and then practising being out of control.

Ten to twenty minutes at a time. Stop after repeated large oscillations or frustration and come back later. High-quality repetitions beat an hour of survival behind a tanker, and the community figure for this skill is about thirty minutes a day for two weeks. It feels impossible until it abruptly does not.

End on something you did right — a stable pre-contact, one clean plug, a controlled disconnect. And a reset is never a failed run: backing out deliberately is the procedure, not a surrender.

What you need: the F-16C-50 · Viper, and nothing else. Every ride in THIS copy of the syllabus is flown in it, behind a KC-135 Stratotanker (boom) at **300 KIAS, 22,000 ft**, on free Caucasus in the Modern (2000s+) era. No mods, no scripts, no external tools.

That combination was your choice at download, and this document was generated for it — the sight pictures, the track speed and the kneeboard cards below all describe the aircraft you actually picked rather than a default somebody else flew.

What is graded, and what is not

10 of the 11 rides are graded **inside the mission**, using Mission Editor triggers. That grader measures whether you are in the close-in envelope, whether you are on the tanker's speed, how long you hold both at once, and how often you fall out.

It does **not** measure whether you made contact, how long you held the plug, or how much fuel you took. DCS mission triggers cannot see a refuelling event or read your fuel, and doing that would mean shipping a script inside the mission. So the grader reports what it can actually see and the briefs say so on every card. You will not be handed a certificate we did not earn.

The tolerances are deliberately wide. A grader that calls a correct pilot wrong teaches you to ignore it, and after that every call is noise. If you fly it properly you will never be told off; if you are sloppy you may still pass. That trade is on purpose.

The syllabus

Ride numbers match the Academy plan's own 0-9 sequence, so the plan and the Library are talking about the same rides. Ride 10 is an extra: a variation on something the track already taught, not a new demand.

0. Orientation and Control Check [graded · station keeping]

The diagnostic. Can you sit beside a tanker for a minute, and does your throttle do what you think? Three clean close-stop-aft-stop cycles and you may move on.

NEW DEMAND: none. This is the diagnostic. Can you sit next to a large aeroplane and stay there, and do your controls do what you think they do.

1. Formation on the Tanker [graded · station keeping]

Find him, match him, park line abreast. No contact this ride.

NEW DEMAND: a stable formation sight picture, held, with no contact objective at all.

2. Closure Control to Pre-Contact [graded · closure control]

Ten arrivals at the contact position, zero plugs. Two closure numbers — one for the approach, and one three times slower for the last few feet.

NEW DEMAND: arriving slowly enough that stopping is a decision rather than a hope.

3. First Contact on the Boom [graded · position while working]

You start at pre-contact. Take fuel, back out, do it again.

NEW DEMAND: contact mechanics. One plug, then another — the connection itself, not yet the minute after it.

4. Sustained Contact and Fuel Transfer [graded · position while working]

Contact is a moment; this is the minute after it. Forty-five continuous seconds — and you are getting heavier the whole time, so the power that held you at the start will not hold you at the end.

NEW DEMAND: staying there. Contact is a moment; this ride is about the minute after it.

5. Disconnect Recovery and Reset [graded · position while working]

Get it wrong on purpose, ten times, and recover each time. A reset is not a failed run — it is the skill that makes refuelling reliable rather than lucky.

NEW DEMAND: getting it wrong ON PURPOSE, and recovering.

6. Refuelling Through the Turn [graded · position while working]

Hold position through the tanker's turn. Inside you slide forward, outside you fall back and get high, and the sight picture rotates while your reference does not.

NEW DEMAND: staying in position while the tanker banks.

7. Rendezvous, Radio Flow and Join [graded · station keeping]

Everything before the part you have been practising: navigation, the radio flow, the join, observation. The commute is back, once, on purpose — arriving already matched is its own skill.

NEW DEMAND: everything that happens BEFORE the part you have been practising. Navigation, the radio flow, the join, observation, and only then pre-contact.

8. Qualification Check Ride [graded · position, quietly]

The check ride. Grader on, coaching off. Three of the five criteria are measured by the mission and two are on your honour — and the card says exactly which is which.

NEW DEMAND: doing all of it with nobody talking to you.

9. Operational Transfer on a Live Sortie [not graded]

A real sortie with real tasking and a tanker on station. No prompts, no grade — the measure is whether you came home with the job done.

NEW DEMAND: mission workload and a fuel decision — refuelling as something you fit around a job, rather than the job.

10. Night Refuelling (extra) [graded · position, quietly]

Same procedure, no horizon, and the cues you were using are gone.

NEW DEMAND: none. Same procedure with the visual cues removed, which is why it is an extra and not a ride.

Where the numbers come from

Four different kinds of claim appear in these rides and they are not printed as though they were one. The stabilised-then-cleared shape at pre-contact is doctrine (ATP-56). The 1-3 kt approach closure is quoted (Stephenson, *Air Refueling Receiver*). The roughly one foot per second in the last few feet is quoted (KC-46 programme literature). The 15-second gate and the 60-second standard are **ours** — no document we hold states a duration, and the guide says so rather than implying a manual.

The tanker's speed and altitude are a tuned operating point, not a quotation: published real-world bands are wide and depend on receiver, weight and altitude. What is in the mission is a setting chosen so the receiver can actually hold contact in DCS, sitting inside the real band.

Your tanker, and the ones you were not offered

KC-135 Stratotanker (boom) — flying **300 KIAS at 22,000 ft** for this receiver. The USAF standard. You fly formation on the tanker and the boom operator flies the boom to you.

Why the track is higher than the book figure. A KC-135 Stratotanker would normally work at 20,000 ft. This map's highest ground is about 18,510 ft and you begin every ride 1,000 ft **BELOW** the track, so the track is raised to 22,000 ft to keep the LOW aeroplane — you — above it. Thinner air also means your indicated airspeed buys less manoeuvring margin: fly smaller corrections.

Kneeboard card — the procedure

This is the card that ships on your kneeboard in every ride of this track. It is reproduced here so you can print it.

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AIR REFUELLING — KC-135 Stratotanker (boom)

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TRACK: 300 KIAS at 22,000 ft

TYPE: FLYING BOOM — the boom operator flies the boom to you

THE FOUR POSITIONS. You are always in exactly one of them.

1. RENDEZVOUS — behind and BELOW, 1 nm.
Match 300 KIAS before you are close, not after. Closure you have not noticed is closure you cannot stop. Below the track, always: if it goes wrong you descend away, and down is safe.
2. OBSERVATION — line abreast the wing, stepped down.
Stabilise here. If you cannot hold this position hands-steady for thirty seconds you are not ready for the next one, and going anyway is how people hit tankers.
3. PRE-CONTACT — astern, 10-20 ft back, boom stowed.
This is the position that decides the sortie. Everything from here is FORMATION FLYING, not aiming.
4. CONTACT.
Fly formation on the TANKER — not on the boom. Pick a reference on the tanker's belly or the pod and hold it still. The operator will fly the boom into your receptacle; your only job is to stop moving. Director lights or the operator's calls will walk you fore and aft.

BEFORE YOU FLY THIS: CHECK YOUR STICK.

The most common fix for 'I cannot refuel' is not technique, it is a control curve. A plastic ball gimbal has centre play; a mechanical one does not; a longer stick is a mechanical curve already. So there is NO right number and anyone who gives you one is guessing about your hardware. The principle: add pitch and roll curve until small corrections stop overshooting, and no further — curve buys you precision at centre and sells you precision at the edges. Deadzone is a defect compensator, not a technique aid: set it just past your stick's measured centre noise, usually 0-3, and leave it there. Do NOT curve the throttle. Anticipate instead — the spool is the lag.

WHY IT OSCILLATES, WHICH IS NOT A CHARACTER FLAW.

What you are doing when you chase the basket has a name: pilot-induced oscillation. You correct for where you ARE instead of where you are GOING; the correction arrives after a lag; it overshoots; you correct harder. Push it far enough and your inputs land 180 degrees out of phase with the aeroplane and the oscillation grows instead of damping. Human reaction time puts the worst band at roughly one correction a second — which is exactly the rate people correct at when they are tense.

That reframes the whole problem. You do not need to be better. You need LESS GAIN, and there are four levers:

1. Relax your grip. Tension is gain you did not ask for.
2. One input, then wait. Waiting is a technique, not hesitation.
3. Look at the tanker, not the boom or basket. The basket has its own oscillation — watching it puts a second oscillator inside your control loop. This is not a comfort tip; it is removing a noise source.
4. Curve the axis. That is gain reduction in software.

The F-16 does lever 4 for you in hardware: opening the AR door drops the flight-control gains. The jet agrees with the diagnosis.

THE THREE MISTAKES, in the order people make them:

- CHASING. You correct, it overshoots, you correct harder. Stop correcting. Freeze the stick, let it settle, then make ONE small input.
- STARING. Looking at the boom or the basket makes you fly it. Look at the TANKER, wide, the way you look at a lead in formation.
- THROTTLE STEPS. Big handfuls of power arrive late and leave late. Think in single percent, and lead the correction.

IF IT IS NOT WORKING: back out to pre-contact and breathe. Nobody has

ever fixed a bad approach by pressing it. The tanker is not going anywhere.

WHY THE TRACK IS HIGH: this map's highest ground is about 18,510 ft, and you start the sortie 1,000 ft BELOW the track. The track sits at 22,000 so the LOW aeroplane – you – still clears it. A refuelling track you can fly into a mountain is not a training aid.

WHAT GOOD LOOKS LIKE – pass criteria, so you can stop guessing:

- PRE-CONTACT, THE GATE: stabilised with zero closure for 15 seconds. That is the minimum before you move forward, and it is the doctrinal shape of the clearance – stabilised first, then cleared. Fifteen seconds is a gate, not a score.
- PRE-CONTACT, THE STANDARD: hold that same position for 60 seconds without a correction you did not intend. That is not the gate and it is not quoted from anyone – it is OUR proficiency mark, set where a pilot who can do it will not be surprised by anything in contact.
- CLOSURE, APPROACH: 1 to 3 knots over the tanker. Not 10. At 300 KIAS that is a needle's width. (One knot is about 1.7 feet per second, which is the unit the next line is in.)
- CLOSURE, LAST FEW FEET: bleed it to about ONE FOOT PER SECOND – roughly half a knot, slower than the approach by a factor of three. This is the number people miss: the closure that gets you to the position is not the closure that makes the connection.
- IN CONTACT: hold roughly a three-foot box. Small.
- PROGRESSION: from 1-3 second connections, to a plug you can hold, to a full transfer. Count them and write the number down.

HOW LONG THIS TAKES, honestly: about thirty minutes a day for two weeks. That is the figure the community converges on, and marathon sessions make it worse, not better. It feels impossible until it abruptly does not. You are not the exception.

The USAF standard. You fly formation on the tanker and the boom operator flies the boom to you.

Kneeboard card — what the grader sees

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THIS RIDE IS GRADED — and here is exactly what by

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FOCUS: position while working.

NOTHING IS SCORED FOR THE FIRST 20 SECONDS. That window is yours: in VR, recentre sitting naturally with square shoulders and set seat height so the tanker reference is visible without a crouch you cannot hold for ten minutes. On a monitor, use it to find him and settle. A sight picture measured before you have set your viewpoint is measuring the headset, not the pilot.

MEASURED:

- Are you inside 150 m of the tanker (the close-in envelope: observation, pre-contact, contact).
- Are you within 10 kt of his track speed.
- How long you hold both at once. 15 s is the gate, 60 s is the standard.
- How many times you fall out after getting there.

NOT MEASURED, and no card in this product will pretend otherwise:

- Whether you made contact.
- How long you held the plug.
- How much fuel you took.

DCS mission triggers cannot see a refuelling event or read your fuel. Anything that claims to grade your plug without running a script is guessing, and we would rather tell you what we can actually see.

The tolerances are deliberately WIDE. A grader that calls a correct pilot wrong teaches you to ignore it, and then every later call is noise. If you are flying it properly you will never be told off; if you are sloppy you may still pass. That trade is on purpose.

COACHING FADES. The closure call fires at most three times and then stops. After three you know. Everything else is on the F10 menu, when YOU want it — asked-for feedback gets used, unasked-for gets tuned out.

The position indicator — every state, at size

This is the panel that appears on the left of your screen during the coached rides. It is printed here at full size so you can learn it on the ground rather than at the boom — the whole point of it is that you recognise a state without reading it.



Ball against the bars = your height. **Chevrons** = fore/aft: up is closing, down is falling back. **Green HOLD** is the only state not asking you for something. **Red POWER OFF** is the safety call — act on that one immediately.

Kneeboard card — the position indicator

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THE POSITION INDICATOR – left of your screen, not the corner

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A small panel on the LEFT, level with your eye line – roughly where a naval aviator already looks for the ball. Read it with the edge of your vision; you should never have to leave the tanker to use it.

HOW TO READ IT, and it is one glance:

- THE BALL against the two datum bars is your HEIGHT. Above the bars you are high, below them you are low, between them you are within 200 ft of the track. Same idea as the meatball, same glance.
- THE CHEVRONS are FORE/AFT. Pointing up at the top: you are CLOSING. Pointing down at the bottom: you are FALLING BACK. No chevrons: you are within 5 kt of his speed.
- THE WORD backs up the fore/aft state, because that is the axis your throttle has to act on. FAST means overtaking, SLOW means dropping back. Only when both axes are clean does it say HOLD, in green.
- A RED PANEL WITH THREE ARROWS and POWER OFF is the safety state: more than 25 kt of overtake inside a quarter mile. Act on that one immediately.

IT UPDATES AT MOST ONCE EVERY 4 SECONDS, deliberately. An indicator that redrew the instant anything changed would strobe every time you sat on a band boundary, which is most of the time. The cost is that a real change can take a moment to appear – worth it, because a flickering panel is one you learn to ignore.

IT DOES NOT SHOW LEFT/RIGHT, and that is deliberate rather than unfinished. The mission knows how far you are from the tanker, not which side you are on – the check is a sphere. We could draw a lateral arrow and it would be guessing, and a training aid that invents a cue is worse than one that admits a gap. Your lateral position comes from the sight picture, the way it does in the jet.

Colour is never the only cue: every state is also a shape, a position and a word, so it reads the same if you are colour-blind or the contrast is poor.

TURN IT OFF from the F10 menu whenever you like. It does not appear at all on the qualification ride – that one is silent by design.

Kneeboard card — your hardware

Read once, on the ground. Most of what stops people refuelling is set up out here, not flown in there.

AIR REFUELLING — YOUR HARDWARE

Read once, on the ground. Most of what stops people refuelling is set up out here, not flown in there.

--- THE THROTTLE MATTERS MORE THAN THE STICK -----

Under-appreciated, and consistent across every source: the bottleneck is usually throttle RESOLUTION, not stick precision. A separate throttle unit wins on travel alone — a twist-stick slider cannot give you the movement range to make a one-percent change.

- Think in taps. 'A slight tap forward', 'half a tap back'.
- On a Warthog-style throttle, tune out the afterburner detent bump near 90 % — it is a sensitivity discontinuity in the middle of the task.
- Speedbrake moves the engine into a more responsive RPM band. Free throttle authority — but it also changes drag and, in some aircraft, pitch. Use it where your aircraft's technique supports it. It is not a universal beginner trick.
- Expect to add power as fuel transfers. You are getting heavier.

--- PULSE AND COUNTER-PULSE: HOW THE THROTTLE IS ACTUALLY FLOWN -----

The stick puts you in the right vertical and lateral picture. The throttle controls the fore/aft TREND, and it is the part almost nobody is taught.

First find your BASELINE: the power setting that produces zero fore/aft drift once you are matched. Everything else is a movement away from it and back.

1. Drifting aft? Add a small pulse of power.
2. BEFORE the closure becomes obvious, take most of it back out. That is the counter-pulse, and skipping it is the whole problem.
3. Watch the new trend. Make the next small correction.
4. Reverse the pattern for forward drift.

Pilots call it walking the throttle. The goal is NOT a motionless throttle — it is a series of small, early corrections around a baseline you know.

WHY EARLY MATTERS, in numbers: one knot of mismatch is about 1.7 feet per second. A difference too small to see becomes an aircraft length in ten seconds, and by the time the drift is obvious the engine lag means you need a big correction — which then leaves you accelerating the other way.

THE FIVE THROTTLE ERRORS:

- Waiting for a large error before correcting.
- Leaving the correction in, so you sail through zero into an overshoot.
- Chasing indicated airspeed. The number is a cross-check; relative motion against the tanker is the control problem.
- Large alternating movements — fore/aft PIO with the stick perfectly steady.
- Freezing a memorised setting during transfer. You are getting heavier; the baseline moves.

THE DRILL, before any contact attempt. From a safe in-trail position: close 20-30 ft, stop the closure without changing your vertical or lateral picture, drift aft 20-30 ft, stop that, return to the original picture. THREE smooth cycles and you are ready for pre-contact. It teaches engine response and counter-pulsing with nothing at stake.

ON THROTTLE CURVES: start LINEAR and stay there. Eagle Dynamics' own controller guidance treats linear as the realistic baseline and curves as compensation for gaming-hardware limits — and warns that softening the centre amplifies input everywhere else. Add a throttle curve only if a short throw or a detent genuinely compresses the AAR power band on YOUR hardware, and never because you copied someone else's. Throttle length, detents, friction, engine model, tanker speed and stores all change which part of the axis you are actually using.

FRICTION: set it so the throttle stays where you released it but still

moves from the fingers without a sticky breakout. Anchor your forearm or the heel of your hand if the hardware lets you, and move from the fingers.

--- FEET OFF THE PEDALS -----
One instruction, no dissent found anywhere: keep your feet completely off the rudder pedals at the boom. Unintended yaw is a correction you did not know you made.

--- FORCE FEEDBACK: WE LOOKED, AND THERE IS NO EVIDENCE -----
You would expect a direct-drive FFB base to help – no cam breakout at centre, so micro-inputs should be cleaner. Nobody has written down that it does. Searching the ED forums, the VPforce documentation and the review sites turns up long discussions of FFB for helicopters, warbirds and stall buffet, and essentially nothing on refuelling or formation. The one thread asking the question directly ended with the community diagnosing the pilot's TANKER SPEED, not his stick.

ONE HOLE IN THAT SEARCH, and it is a real one: we could not reach REDDIT. r/hoggit is one of the largest DCS venues and none of it was readable from where this was researched. So the honest claim is 'no evidence in everything we could search', not 'no evidence anywhere'. If you have seen a good r/hoggit thread on FFB and refuelling, that is genuinely new information and we would rather know.

So: if you have one, good. If you are considering buying one to fix your refuelling, that is not a supported reason.

BUT IF YOU DO HAVE FFB, ONE THING ON THIS CARD IS WRONG FOR YOU: the advice to add an axis curve. VPforce's own documentation says curves and saturation are incompatible with FFB and must be disabled. The reason is structural – on a spring stick DCS only READS position, so a curve is a harmless remap; on FFB, DCS also WRITES position to represent the trim point, and a curve desynchronises the two.

Use these instead:

- SPRING GRADIENT up. More force per degree means a smaller displacement for the same hand force. This is the honest substitute for a curve.
- DAMPING up. Resistance proportional to how fast you are moving the stick – so it suppresses the fast, panicky input and leaves the slow deliberate one alone. That is rate-dependent gain reduction, which is arguably better than a curve rather than merely different.
- NOT friction, and NOT inertia. Friction adds a breakout you have to overcome, which recreates the step-input problem you were escaping. Inertia adds lag to reversals, and reversals are the whole task. (That last paragraph is reasoning from the effect definitions, not a quoted source. Nobody has published AAR-specific FFB settings.)

FFB TRIM GOTCHA, and it is a real one: your stick physically moves to the trim point. Trim BEFORE you go to pre-contact, never at the boom – otherwise the stick relocates under a hand that is trying to hold it still. Modules with genuine per-module FFB are the F-14, F-4E and Mirage F1; the F-16 and F/A-18 have little beyond trim.

--- VR: IT HELPS, BUT NOT WHERE YOU THINK -----
VR is genuinely better for this, and the mechanism is stereo depth – which has a range limit worth knowing, because depth resolution falls off as the SQUARE of distance.

The US Navy's own simulator requirement asks for accurate depth judgement between 5 and 100 ft. That is the number to hold on to:

- Inside ~100 ft, stereo is doing real work. At 20 ft it can resolve a few centimetres.
- At 1 nm astern it resolves kilometres. It is contributing NOTHING.

Which means: VR does not help you find the tanker or fly the rejoin. Out there you are using angular size, perspective and closure rate, and those are all available on a flat screen. VR helps in the last hundred feet, which happens to be the part that is hard.

This is not just a sim opinion. The KC-46's Remote Vision System – real boom operators working from camera displays instead of a window – has spent a decade failing on exactly this, and a study of stereo displays for it found that adding stereo and hyper-stereo improved refuelling performance.

VR TIPS THAT ARE ACTUALLY SPECIFIC TO VR:

- HOLD YOUR HEAD STILL at contact. Head movement is its own input. Flat-screen pilots have been known to switch head tracking OFF for the plug; you cannot, so the discipline replaces the switch.
- DO NOT ZOOM at the boom. Zoom changes the projection and corrupts the stereo cue you came to VR for. Use it to read the tanker's lights, then drop it before you move forward.
- The DCS VR 'IPD' setting is WORLD SCALE, not your eye spacing. Community values run 45-55, calibrated by lining your real shoulder and elbow up with the virtual pilot's. Worth knowing that a larger value is effectively hyper-stereo – the thing that improved the KC-46 results. Whether that helps at the boom is UNTESTED; if you try it, change one thing at a time.
- Approaching at about 45 degrees off gives better closure perception in VR than coming straight up the tail.
- The RIGHT basket is harder, and not because of you: you are looking away from the tanker's lights and structure, so your references go.

SET UP BEFORE YOU FLY, NOT AT THE BOOM:

- Get your headset's PHYSICAL lens spacing right and find the optical sweet spot. That is a different setting from the DCS world-scale value above, and getting the physical one wrong makes every sight picture slightly untrue in a way you will blame on your flying.
- Bind RECENTRE VR HEADSET, and recentre sitting naturally with square shoulders looking straight ahead. Then set virtual seat height so the tanker reference, boom markings or hose pod stay visible without a crouch. Use the seat adjustment, not your neck – you cannot hold a duck for ten minutes.
- Bind EVERYTHING you need from observation through disconnect. Taking a hand off the HOTAS to hunt for the mouse is bad anywhere and much worse blind.
- Favour steady frame timing and a clear tanker silhouette over visual extras. VR pixel density is expensive, and stuttering at contact is a flying problem, not a graphics preference.

IF A TUTORIAL'S SIGHT PICTURE DOES NOT WORK FOR YOU, do not go straight to your stick curves. Check the things that actually differ first: your virtual eye height, fore/aft seat position, which tanker it was, which hose station, and your headset scale. A picture recorded from a different viewpoint is not a picture of your problem.

AND IF TRACKING DROPS, the image stutters, or your eyes start to ache – back out. Twenty feet from a tanker is the wrong place to troubleshoot a headset.

ONE KNOWN DCS DEFECT, so you stop blaming your eyes: the KC-135's director lights ARE NOT LIGHTS. They are a pre-rendered texture – an ED beta tester confirmed it and the file is in the KC-135 texture pack. In VR and on large displays they are hard to read, it has been reported since 2021, and it is still open. A community texture mod exists that brightens them. (For what it is worth, a real KC-135 receiver instructor says they are difficult to see in the aeroplane too.)

--- FLAT SCREEN -----
Set your field of view GEOMETRICALLY CORRECT before you practise this. On a flat screen your only closure cue is angular size growth, so a wrong FOV systematically miscalibrates the one thing you are trying to judge. Rule of thumb: when your monitor's width equals your viewing distance, correct FOV is about 60 degrees; scale from there.